As discussed above, the intuition for this result is straightforward: for any permutation other than the identity permutation, there exists at least one element which must be moved from a later position to an earlier position in the sequence. We'll argue, by Lemma 1, that the causal attention mask precludes this.

Proof. Consider some P that is not the identity matrix. Recall that rows and columns of P must sum to one. Let P_i denote the ith-row of P. We first prove that P must have at least one nonzero below-diagonal entry. That is, we claim that there exists i such that $P_i = e_j$ for $j < i$. To see this, let i be the index of the last row which is not equal to its corresponding elementary basis vector e_i. (By the fact that P is not the identity, there must some such row index.) Since for all $j > i$, the jth row is equal to e_j, the last $d - i$ entries of P_i must be zero. Also, since $P_i \neq e_i$, the diagonal entry must be zero. Therefore, the non-zero entry of P_i must be at some entry j for $j < i$. This is a non-zero below-diagonal entry of P_i. Put another way: any non-identity permutation has at least one "cycle" ($i_1 \to i_2 \to \cdots \to i_1$), which implies there exists a nonzero below-diagonal entry.

Now, let i, j be any indices such that $P_i = e_j$ for $j < i$. In order to store Y in some block of the residual stream, the jth row of Y cannot be output into any row of the residual stream more than $d - j$ rows from the bottom. Intuitively, if Y is to "fit" into the residual stream, row j cannot be output at too low of a row in the residual stream: there must be space for at least $d - j$ more rows. Formally, the jth row of Y must be output somewhere in the ℓ-th row of the residual stream for $\ell \leq d + j + 1$. However, since $Y_P = PY$ and $j < i$, Y_j is located in the ith row of Y_P, which is in row $d + i + 1$ of the input. Since $d + i + 1 > d + j + 1 \geq \ell$, by Lemma 1, row ℓ of the residual stream must be invariant to row $d + i + 1$. Thus, for any fixed weight matrices $\{A^{(m)}\}_{m \in [k]}$, there exists some Y such that it is not output to the residual stream. □

C.2 Proof of Theorem 2

Theorem 2. *There exists parameter matrices $A^{(1)}$ and $A^{(2)}$ such that, for any permutation matrix P and target matrix Y, a two-layer, decoder-only, attention-only, causal mask-free, transformer parameterized by $A^{(1)}, A^{(2)}$ given $[P; Y_P]$ as input outputs Y to a block of the residual stream.*

Proof of Theorem 2. We proceed by construction.

The first layer copies P into the bottom block-row of the residual stream. For constant β_1, we will have attention weights represented by the following block matrix

$$A^{(1)} = \beta_1 \begin{bmatrix} 0_d & 0_d & 0_d \\ 0_d & 0_d & 0_d \\ 0_d & I_d & 0_d \end{bmatrix}$$

where each 0_d is the $d \times d$ all-zeros matrix and $I_d \in \mathbb{R}^{d \times d}$ is the $d \times d$ identity matrix. This implies pre-activations

$$XA^{(1)}X^\top = \beta_1 \begin{bmatrix} 0_d & 0_d \\ I_d & 0_d \end{bmatrix} \in \mathbb{R}^{T \times T}$$

This implies activations

$$\mathcal{S}(XA^{(1)}X^\top) = \begin{bmatrix} 1/(2d)\mathbf{1}\mathbf{1}^\top & 1/(2d)\mathbf{1}_d\mathbf{1}_d^\top \\ I_d & 0_d \end{bmatrix}$$

where $1/(2d)\mathbf{1}\mathbf{1}^\top$ is the $d \times d$ constant matrix with entries $1/(2d)$. Finally, we have layer-1 outputs

$$\mathcal{S}(XA^{(1)}X^\top)X = \begin{bmatrix} 1/(2d)\mathbf{1}\mathbf{1}^\top(P+Y_P) & \dots & \dots \\ P & \dots & \dots \end{bmatrix}$$

where we write $\dots$ for block matrices that will not factor into later computations and therefore do not matter.

The second attention layer computes Y. Recall that the input to the second attention is

$$\begin{aligned} h^{(1)} &= \begin{bmatrix} X & \mathcal{S}(XA^{(1)}X^\top)X \end{bmatrix} \\ &= \begin{bmatrix} P & I_d & 0 & 1/(2d)\mathbf{1}\mathbf{1}^\top(P+Y_P) & \dots & \dots \\ Y_P & 0 & I_d & P & \dots & \dots \end{bmatrix} \end{aligned}$$

Define a constant β_2. We will have attention weights

$$A^{(2)} = \beta_2 \left[\begin{array}{ccc|ccc} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

This implies pre-activations

$$\begin{aligned} h^{(1)}A^{(2)}h^{(1)\top} &= \beta_2 \begin{bmatrix} I_d \\ 0_d \end{bmatrix} \begin{bmatrix} 1/(2d)\mathbf{1}\mathbf{1}^\top(P+Y_P)^\top & P^\top \end{bmatrix} \\ &= \beta_2 \begin{bmatrix} 1/(2d)\mathbf{1}\mathbf{1}^\top(P+Y_P)^\top & P^\top \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Note that $1/(2d)\mathbf{1}\mathbf{1}^\top(P+Y_P)^\top < 1$ so, as $\beta_2 \to \infty$, we have activations

$$\mathcal{S}(h^{(1)}A^{(2)}h^{(1)\top}) = \begin{bmatrix} 0 & P^\top \\ 1/(2d)\mathbf{1}\mathbf{1}^\top & 1/(2d)\mathbf{1}\mathbf{1}^\top \end{bmatrix}$$

This yields layer-2 outputs

$$\mathcal{S}(h^{(1)}A^{(2)}h^{(1)\top})h^{(1)} = \begin{bmatrix} \dots & \dots & \dots & \dots & \dots & \dots \\ P^\top Y_P & \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

Finally, notice $P^\top Y_P = Y$ so that block recovers the desired canonical order. □

C.3 Proof of Theorem 3

Theorem 3. *There exist parameter matrices $A^{(1)}$ and $A^{(2)}$ such that, for any permutation matrix P and target matrix Y, a two-layer decoder-only attention-only transformer parameterized by $A^{(1)}, A^{(2)}$ given $[P; Y_P; S]$ as input outputs Y to a block of the residual stream.*

Proof of Theorem 3. Observe, since we pad the beginning of the input with a `BOS` token to be all zeros, that

$$
\begin{aligned}
X &= \texttt{embed}(s_{1:T}) \\
&= \begin{bmatrix} 0 & 0_{1\times T} \\ P & \\ Y_P & I_T \\ S & \end{bmatrix} \\
&= \begin{bmatrix} 0_{1\times d} & 0_{1\times d} & 0_{1\times d} & 0_{1\times d} \\ P & I_d & 0_d & 0_d \\ Y_P & 0_d & I_d & 0_d \\ S & 0_d & 0_d & I_d \end{bmatrix} \in \mathbb{R}^{(T+1)\times(d+T)}
\end{aligned}
$$

The first layer copies P into the third block-row of the residual stream. For constant β_1, we will have attention weights represented by the following block matrix

$$
A^{(1)} = \beta_1 \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & I_d & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}
$$

where each 0 is the $d \times d$ all-zeros matrix. This implies pre-activations

$$
XA^{(1)}X^\top = \beta_1 \begin{bmatrix} 0 & 0_{1\times d} & 0_{1\times d} & 0_{1\times d} \\ 0_{d\times 1} & 0_d & 0_d & 0_d \\ 0_{d\times 1} & I_d & 0_d & 0_d \\ 0_{d\times 1} & 0_d & 0_d & 0_d \end{bmatrix} \in \mathbb{R}^{(T+1)\times(T+1)}
$$

Let $\Delta^{(1)}$ be the lower triangular matrix with $\Delta_{ij} = 1/i$ for $j \leq i$. Let $\Delta^{(1)}$ be the $d+1 \times d+1$ matrix with $\Delta_{ij} = 1/(2d+i+1)$. And as $\beta_1 \to \infty$,

$$
\mathcal{S}(\text{MASK}(XA^{(1)}X^\top)) = \begin{bmatrix} \Delta^{(1)} & & 0 & 0 \\ 0_{d\times 1} & I_d & 0 & 0 \\ 0_{d\times 1} & \Delta^{(2)} & \cdots & \cdots \end{bmatrix}
$$

We will write dots when the values in a block do not matter for subsequent computations. This yields layer 1 outputs:

$$
\mathcal{S}(\text{MASK}(XA^{(1)}X^\top))X = \begin{bmatrix} \Delta^{(1)}[0_{1\times d}; P] & 0_{1\times d} & 0_{1\times d} & 0_{1\times d} \\ \cdots & \cdots & \cdots & \cdots \\ P & 0_{d\times d} & 0_{d\times d} & 0_{d\times d} \\ 0_d & \cdots & \cdots & \cdots \end{bmatrix} \in \mathbb{R}^{T\times(d+T)}
$$

The second layer will produce $P^\top$ in the activations so that Y is written to the residual stream. Recall:

$$
h^{(1)} = \begin{bmatrix} 0_{1\times d} & 0_{1\times d} & 0_{1\times d} & 0_{1\times d} & \Delta^{(1)}[0_{1\times d}; P] & 0_{1\times d} & 0_{1\times d} & 0_{1\times d} \\ P & I_d & 0_d & 0_d & \cdots & \cdots & \cdots & \cdots \\ Y_P & 0_d & I_d & 0_d & P & 0_{d\times d} & 0_{d\times d} & 0_{d\times d} \\ 0_d & 0_d & 0_d & I_d & 0_d & \cdots & \cdots & \cdots \end{bmatrix}
$$